

Nguyen Luong Gia Bao

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PARTICULARS

EDUCATION

University of Economics and Law – Vietnam National University

2023 – 2027

Bachelor of Financial Technology

GPA: 3.46/4.0 (8.26/10)

RESEARCH INTERESTS

- **Financial Behavior & Empirical Finance:** causal inference, econometric methods, and event-study analysis
- **Financial Engineering & Applied AI:** derivative pricing, risk management, deep learning, reinforcement learning, and graph neural networks
- **Algorithmic Game Theory & Market Design:** equilibrium computation in continuous double auctions, mechanism design, and multi-agent learning with realistic market frictions

DISSERTATION

Title: “State-Dependent Graph Portfolio Optimization using Selective State Space Models and Regime-Adaptive Graph Learning”

Advisor: MSc. Ngo Phu Thanh

My dissertation examines regime-dependent financial networks and their role in portfolio optimization. The study develops a selective state space framework to model asset-return dynamics across changing market regimes, while a regime-adaptive graph learning mechanism captures time-varying cross-asset dependencies. The resulting state-dependent graph representation is incorporated into a portfolio construction pipeline that adapts to market conditions and seeks stronger risk-adjusted performance during periods of heightened volatility and regime transition.

RESEARCH EXPERIENCE

51Labs – Researcher

August 2026 – Present

Research hub of Miquant, Ho Chi Minh City, Vietnam

- Leading research projects and formulating well-defined research questions in AI and quantitative finance.
- Designing research methodologies, constructing datasets, developing quantitative and machine learning models, and establishing appropriate empirical validation frameworks.
- Conducting experiments, evaluating model robustness, interpreting quantitative results, and refining models based on empirical evidence.
- Taking primary responsibility for drafting, revising, and managing research manuscripts, ensuring methodological consistency and clear presentation of findings.
- Presenting research results for internal review, incorporating technical feedback, and driving subsequent stages of research development.

National Yang Ming Chiao Tung University

January 2027 – June 2027 (Accepted)

Hsinchu, Taiwan

- Research Intern at the Economics and Computing Laboratory – PI: Prof. Dr. Po-An Chen, Director of Institute of Information Management.
- Investigating explainable AI methods for transparent and interpretable decision systems in economics and finance.
- Exploring algorithmic game theory and theoretical machine learning for strategic decision-making and multi-agent optimization.
- Assisting an ongoing 3-year NSTC research project (2024–2027) on convergences in multiagent online/reinforcement learning and robust market making; contributing to the extension and integration of results across related ongoing manuscripts.

INDUSTRY EXPERIENCE

Miquant – Quantitative Research Analyst
Ho Chi Minh City, Vietnam

October 2025 – Present

- Conducted quantitative research and analysis for financial markets using advanced statistical methods
- Developed and implemented mathematical models and algorithms for optimized trading strategies
- Analyzed complex market data and designed comprehensive risk management frameworks

Lumination – Data Analyst
Ho Chi Minh City, Vietnam

August 2025 – October 2025

- Built time-series data pipelines for forecasting and anomaly detection across multiple business domains
- Designed interactive dashboards with automated reporting, reducing manual reporting effort by streamlining data workflows
- Applied statistical modeling and feature engineering techniques to extract actionable patterns from large, complex datasets

Vikki Digital Bank – Risk Consultant
Ho Chi Minh City, Vietnam

February 2025 – May 2025

- Developed risk scoring models using statistical and machine learning approaches to quantify credit and operational risk
- Conducted data-driven risk analysis under regulatory constraints, identifying vulnerabilities in digital banking products
- Designed risk management frameworks and delivered strategic recommendations to senior management based on quantitative findings

PUBLICATIONS AND PRESENTATIONS

JOURNAL

1. **Nguyen Luong Gia Bao**, Ha Quoc Hung, Vo Pham Anh Khoa, Ngo Phu Thanh (2025), “Impact of Margin Cuts on Stock Returns and Trading Value: Evidence from the Vietnamese Stock Market,” *Journal of Economics and Finance* (ESCI Q2, IF: 1.5, ABDC B). DOI: 10.1007/s12197-026-09773-x.
2. **Nguyen Luong Gia Bao**, Ngo Phu Thanh (2026), “World Uncertainty and the Marginal Value of Cash: Suggestive Non-linear Evidence from Six ASEAN Technology Markets,” *International Journal of Economics and Business Research* (Scopus Q4, ABDC C) – Accepted. DOI will be updated.

CONFERENCE

1. Ha Quoc Hung, Ngo Phu Thanh, **Nguyen Luong Gia Bao** (2026), “Market-Relative Temporal Representations for Cross-Sectional Equity Ranking,” *The 10th Vietnam International Conference in Finance (VICIF 2026)* (Presented Paper).
2. **Nguyen Luong Gia Bao**, Ngo Phu Thanh (2026), “World Uncertainty and the Marginal Value of Cash: Suggestive Non-linear Evidence from Six ASEAN Technology Markets,” *7th Vietnam Economist Annual Meeting 2026 (VEAM 2026)* (Presented Paper).
3. **Nguyen Luong Gia Bao**, Ngo Phu Thanh (2026), “State-Dependent Graph Portfolio Optimization using Selective State Space Models and Regime-Adaptive Graph Learning,” *Conference on Optimization, Modeling, Simulation, and Analytics (COMOSA 2026)* (Presented Paper).
4. Tam Phan Huy, Vu Tran Huu Anh, Minh Nguyen Hoang, Phuc Huynh Hoang, **Nguyen Luong Gia Bao**, Trieu Nguyen Hoang (2025), “Causality Analysis of Economic Policy Uncertainty and Volatility in Clean Energy Stock Indexes,” *University of Economics and Law – Student Research Award*.
5. Tam Phan Huy, Vu Tran Huu Anh, **Nguyen Luong Gia Bao**, Tien Vo Quoc, Thu Truong Ngoc Kim, Lien Huynh Mai (2025), “A Multi-Layer Machine Learning and Reinforcement Learning Framework for Portfolio Optimization in VN-100,” *University of Economics and Law – Student Research Award*.

WORK-IN-PROGRESS

SCREENING

1. Ha Quoc Hung, Ngo Phu Thanh, **Nguyen Luong Gia Bao** (2026), “Market-Relative Temporal Representations for Cross-Sectional Equity Ranking,” *Review of Quantitative Finance and Accounting* (ESCI Q2, IF: 2.2, Springer).
2. Ha Quoc Hung, Ngo Phu Thanh, **Nguyen Luong Gia Bao** (2026), “Period-Aware Analog Retrieval for Financial Risk Forecasting,” *Computational Economics* (SSCI/SCIE Q2, IF: 2.1, Springer).
3. **Nguyen Luong Gia Bao**, Ngo Phu Thanh (2026), “World Uncertainty, Cost of Debt, and Firm Financial Stability: Nonlinear and Heterogeneous Evidence,” *Humanities and Social Sciences Communications* (AHCI Q1, IF: 4.7, Springer).
4. **Nguyen Luong Gia Bao**, Ngo Phu Thanh, Vo Thanh Hung (2026), “Best Response Heuristics under Payoff Frictions in Double Auction Quote Selection,” *International Journal of Financial Engineering* (ESCI, IF: 0.7, World Scientific).
5. Nguyen Anh Phong, **Nguyen Luong Gia Bao**, Ngo Phu Thanh (2026), “Stability Pool Allocation and Redemption Routing in Liquity V2: Protocol Reconstruction and Bounded Behavioral Evidence,” *International Journal of Information Technology & Decision Making* (SCIE Q2, IF: 1.9, World Scientific).
6. Nguyen Anh Phong, **Nguyen Luong Gia Bao**, Ngo Phu Thanh (2026), “Learning Objectives and Information Value in Cross-Sectional Stock Ranking: Evidence from the Vietnam Stock Market,” *Annals of Data Science* (Q1, Springer).
7. Ngo Phu Thanh, **Nguyen Luong Gia Bao**, Xuan Thien Nguyen (2026), “High-Frequency Limit-Order Submission Around the 2021 Meta Outage: A Localized Activity Burst Without Broad Directional Coordination,” *Journal of Open Innovation: Technology, Market, and Complexity* (Q1, Elsevier).

UNDER REVIEW

1. Ha Quoc Hung, Ngo Phu Thanh, **Nguyen Luong Gia Bao** (2026), “MambaLOB: Efficient and Uncertainty-Aware Limit Order Book Forecasting for Cryptocurrency Markets,” *International Journal of Information Management Data Insights* (Scopus Q1).
2. **Nguyen Luong Gia Bao**, Ngo Phu Thanh (2026), “State-Dependent Graph Portfolio Optimization using Selective State Space Models and Regime-Adaptive Graph Learning,” *Int. J. of Computational Economics and Econometrics* (ESCI Q4, ABDC C).
3. **Nguyen Luong Gia Bao**, Nguyen Phat Dat, Ngo Phu Thanh (2026), “Double Machine Learning Reveals State Dependent Market Breadth Effects on Equity Returns,” *Discover Sustainability* (ESCI Q1, IF: 5.2, Springer).
4. **Nguyen Luong Gia Bao**, Ngo Phu Thanh (2026), “Economic Policy Uncertainty, Corporate Cash Buffers, and Firm Financial Stability: Double Machine Learning Evidence from Southeast Asia,” *Cogent Economics & Finance* (ESCI Q2, IF: 1.7, ABDC B, Taylor & Francis).
5. **Nguyen Luong Gia Bao**, Ngo Phu Thanh (2026), “Policy Uncertainty, Cash Buffers, and Firm Financial Stability: Evidence from Nonlinear and Machine-Learning Approaches,” *Macroeconomics and Finance in Emerging Market Economies* (ESCI Q3, IF: 1.1, ABDC C, Taylor & Francis).
6. Tran Phuoc, Ngo Phu Thanh, **Nguyen Luong Gia Bao** (2026), “World Uncertainty, Cost of Debt, and Corporate Financial Stability: Panel and Machine-Learning Evidence,” *Research in Economics* (ESCI Q3, IF: 1.1, Elsevier).

ACTIVITIES

AI4I Club

November 2023 – February 2025

External Relations Officer → Marketing and Communications Officer

- Led external relations initiatives and strategic partnership development with industry professionals
- Coordinated collaborative projects with academic institutions and technology companies
- Managed comprehensive marketing campaigns and created engaging content for social media platforms

ACADEMIC HONORS

- Honorable Mention in Scientific Research – University of Economics and Law 2026
- Gold level – WorldQuant BRAIN 2025
- Top 10 – Data Science Competition 2025
- Student of 5 Merits – Faculty of Finance and Banking, University of Economics and Law 2024
- Top 12 – Business Intelligence Competition Season 8 2024
- Top 9 – SAP ERPSIM Student Competition 2024

TECHNICAL PROFICIENCIES

- **Programming:** Python, R, SQL, MATLAB, LaTeX
- **ML & AI:** Deep Learning, Time Series Modeling, Reinforcement Learning, Natural Language Processing
- **Quantitative Finance:** Portfolio Optimization, Risk Modeling, Market Microstructure, Financial Modeling
- **Tools:** PyTorch, TensorFlow, Pandas, NumPy, Jupyter Notebook, Git, VBA, Bloomberg Terminal

LANGUAGES

- **Vietnamese:** Native
- **English:** TOEFL iBT 4.5/6.0 (90)

REFERENCES

Ha Quoc Hung, MSc Founder, Miquant JSC School of Science, Engineering & Technology RMIT University Vietnam hung.ha@miquant.vn	Po-An Chen, Prof. Dr. Director Institute of Information Management National Yang Ming Chiao Tung University poanchen@nycu.edu.tw
Nguyen Anh Phong, Assoc. Prof. Dr. Dean Faculty of Finance and Banking University of Economics and Law, Vietnam National University phongna@uel.edu.vn	Ngo Phu Thanh, MSc Lecturer Faculty of Finance and Banking University of Economics and Law, Vietnam National University thanhnp@uel.edu.vn